

Shreyas Ravi

Innovative mechanical engineer with expertise in automotive systems, vehicle dynamics, structural analysis and mechanical design, specializing in systems design, seeking to drive technological advancements and collaborative engineering solutions. With 6 years of experience in R&D, design and analysis in automotive and mechanical industry with a demonstrated history of team working and leading.

SKILLS

CAD: Catia V5, Solidworks, Creo, Siemens NX, Autodesk

CFD: Ansys Fluent, Star CCM+

Other: LS-dyna, Matlab/Simulink, Adams, Ansys Workbench, EM-tune, C++, Avl vsm; MoTec i2, Ni Multisim, python

Technical Skills: FEM, CAE, Numerical Methods, Structural Mechanics, Automotive Systems, NVH, Model Based Systems Engineering, Agile/Scrum, FMEA, ML, Data Analysis

Professional Skills: Precise & analytical, Goal oriented, Critical thinking, Team work & communication

PROJECTS

- Energy storage-inverter-motor system design for FS vehicle
- CFD analysis of wing & nose of F1 car
- IED Blast simulation on V-hull tank



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DoB:

24 September, 1996

WORK EXPERIENCE

R&D Project Manager Automotive

Rockwool, Netherlands

01/2025 - Present

- **Project Leadership:** Own the end-to-end development of braking solutions, coordinating between commercial units and technical competence centres to align R&D.
- **AI Formulation Tool:** Architected a predictive ML tool for brake pad formulations that reduces development cycles by 50%, transforming material science into a high-value, monetizable commercial asset.
- **Hardware Design & Fluids:** Designed custom sampling hardware utilizing fluid dynamics to ensure laminar flow and consistent capturing of emission particles.
- **Business Intelligence (BI):** Architected BI frameworks to translate complex experimental data into strategic product insights, driving data-informed decisions for formulation and design.
- **Mechanical Analysis:** Conducted deep-dive analysis into stick-slip behaviour and eigenfrequency interactions using ML and regression models to resolve critical NVH and thermal degradation issues.
- **System Optimisation:** Act as the primary technical lead for OEM/Tier 1 customers, delivering system-level optimizations for NVH and particulate constraints.
- **Technical Ownership:** Chair and review technical sessions at FISITA Euro Brake and SAE Brake Colloquium, presenting original research on emission control and NVH strategies.

Doctoral Candidate - Automotive Systems Design

Eindhoven University of Technology, Netherlands

10/2022 - 03/2024

- The program emphasizes technical & professional competencies for designing efficient high-tech automotive systems. Projects at ASML & DAF was undertaken as described in detail below.

Projects undertaken as doctoral candidate

Systems Design Engineer (Thermal) - Tin Mechanics (internship)

ASML, Netherlands

10/2023 - 03/2024

- Conceived & executed virtual simulation models to optimize the performance of a high pressure-high temperature thermodynamic thermal system, resulting in a 15% increase in efficiency & a 20% reduction in energy consumption.

- Exhaust manifold Adaptive Quarter Wave Tube design
- 2020 Imp1 car qualifying and race simulation analysis

ADDITIONAL CERTIFICATIONS

- Model based Automotive Systems Engineering (Chalmers-edX)
- Self-Driving Cars Specialisation (Coursera)
- Business Model Innovation in an Exponential World (TU/e)
- Mechatronics Systems Design (TU/e)

PUBLICATIONS

- "MPC Controller for Autonomous Formula Student Vehicle", SAE Technical Paper 2020-01-0089, 2020, doi: 10.4271/2020- 01-0089
- "Design optimisation of Bicycle Wheel Hub Assembly for Automotive Applications", SAE Technical Paper 2022-01-0262, 2022, doi: 10.42771/2022-01-0262

HOBBIES & INTERESTS

- Professional cricketer KNCB
- Badminton
- 3D Printing
- Adventure sports

LANGUAGES

- **English:** Native/Bi-lingual
- **Dutch:** Beginner (learning)
- **German:** Beginner (learning)
- **Tamil:** Native/Bi-lingual
- **Hindi:** Native/Bi-lingual

Design & Development Engineer (internship)

DAF Trucks, Netherlands

05/2023 - 10/2023

- Developed a range estimation tool for electric trucks using diesel truck data & road gradient insights, achieving up to 5% accuracy without simulations & improving estimation precision by 10-15%.

Sr. Mechanical Engineer (Entrepreneur)

Coexlion, Bangalore

04/2022 - 10/2022

- Created 3D CAD & 2D drawings of structural braces for the rear suspension mounts of the RE motorcycle to mitigate modal harmonic excitations.
- Conducted CAE-FEA analyses on chassis for OEM clients, notably Royal Enfield.
- Utilized 1D modelling & mathematical simulations to analyse kinematic parameter of suspension and steering sub-systems.

Research & Development Mechanical Engineer (Entrepreneur)

InGo Electric, Bangalore

04/2020 - 03/2022

- Led the technical design team of four; Developed kinematic models in Solidworks for mapping out suspension and steering articulation wheel envelopes.
- Created a mathematical 1D Matlab/Simulink model for the motor-CVT setup to develop innovative powertrain system.
- Collaborated with the CAE team to formulate load cases for static & fatigue loading at component & full vehicle levels.
- Handled end-to-end 3D CAD, 2D drawing release, DFMA & supplier communications entailed in development of the vehicle from proto-B stage to mass production phase.

EDUCATION

M.Sc. Motorsport Engineering

Oxford Brookes University, Oxford

09/2018 - 09/2019

- Conducted research on Driverless vehicle analysis and compared control theories for Formula Student Autonomous Vehicle as master's dissertation.

B. Tech Automobile Engineering

SRM University, Chennai

07/2014 - 05/2018

- Designed, analysed & fabricated proof of concept of variable length intake manifold as undergraduate project.

EXTRACURRICULAR ACTIVITIES

R&D Engineer - Team Lead

Oxford Brookes Racing, Oxford

- Led a team as Powertrain EV Lead, managing design, fabrication, & documentation, spearheaded exhaust manifold noise reduction by 3-4 dB while enhancing performance & developed a lateral controller for autonomous vehicles using Simulink.

Team Manager

Infieon Supermileage, Chennai

- Spearheaded a team of 26 individuals, achieving international acclaim for technical innovation at Shell Eco-Marathon Asia '18.